

# CAUSAL MEDIATION ANALYSIS

OTHER CAUSAL MACHINE LEARNING ANALYSES

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# INTRODUCTION

- There is longstanding interest in using data to gain insight into the **mechanism** that underlies the effect of an exposure on an outcome.



- **Mediation analyses** are designed for this purpose.

# EXAMPLES

# MECHANISM IN CLINICAL TRIALS

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### Liraglutide and Cardiovascular Outcomes in Type 2 Diabetes

Steven P. Marso, M.D., Gilbert H. Daniels, M.D., Kirstine Brown-Frandsen, M.D., Peter Kristensen, M.D., E.M.B.A.,  
Johannes F.E. Mann, M.D., Michael A. Nauck, M.D., Steven E. Nissen, M.D., Stuart Pocock, Ph.D.,  
Neil R. Poulter, F.Med.Sci., Lasse S. Ravn, M.D., Ph.D., William M. Steinberg, M.D., Mette Stockner, M.D.,  
Bernard Zinman, M.D., Richard M. Bergenstal, M.D., and John B. Buse, M.D., Ph.D.,  
for the LEADER Steering Committee on behalf of the LEADER Trial Investigators\*

#### LEADER trial

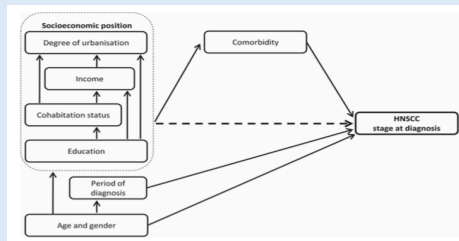
Why does liraglutide reduce the risk of major cardiovascular events?

# HEALTH DISPARITIES RESEARCH

What might explain disparities in health?

## Danish Head and Neck Cancer Group (DAHANCA) database

- people with low socioeconomic position (SEP) more likely have advanced cancer stage at diagnosis;
- interpretation is hindered because people with low SEP more likely have comorbid disease;
- comorbidities make detection of cancer more likely.

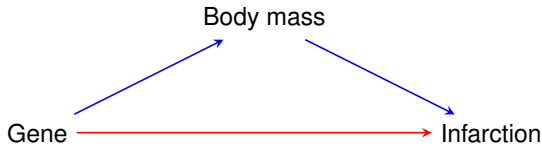


(Olsen et al., 2015)

# PLEIOTROPY IN GENETIC ASSOCIATION STUDIES

Genotypes are often associated with multiple phenotypes.

Is this the result of those phenotypes influencing each other?



## Italian Genetic Study of Early-onset Myocardial Infarction

Does the FTO gene exert an effect on the risk of infarction that is unmediated by changes in body mass?

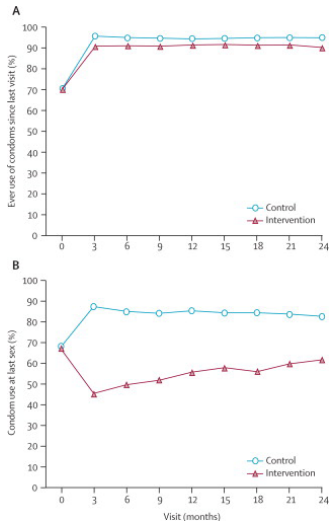
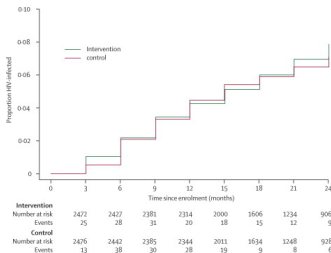
How much of the genetic effect is mediated by body mass?

(Ardissino et al., 2011; Berzuini et al., 2012)

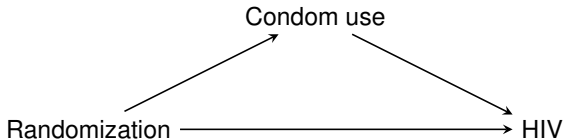
# COMPLICATIONS IN OPEN-LABEL TRIALS (1)

Complications in trials hinder interpretation of the ITT effect.

Padian et al.,  
Lancet 2007



## COMPLICATIONS IN OPEN-LABEL TRIALS (2)



### MIRA trial

What would have been the ITT effect  
had condom use not been affected by the intervention?

(Padian et al., 2007; Rosenblum et al., 2009)

Likewise: complications due to rescue medication.



# TRADITIONAL MEDIATION ANALYSIS

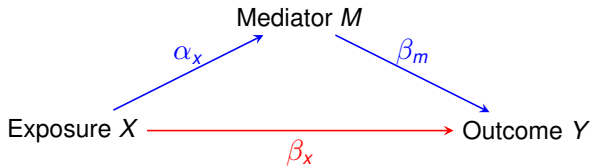
## LOOKING BACK IN TIME...

- Mediation analysis has started to develop around the framework of path analysis and structural equation models.

(Wright, 1934)

- Its development took off in the 80's since the seminal publication by Baron and Kenny (1986) in the psychological / sociological sciences.
- This framework dominates current practice.

# MEDIATION ANALYSIS 1.0



$$Y = \beta_0 + \underbrace{\beta_x}_{\text{direct effect}} X + \underbrace{\beta_m}_{\text{effect of } M \text{ on } Y} M + \epsilon_y$$

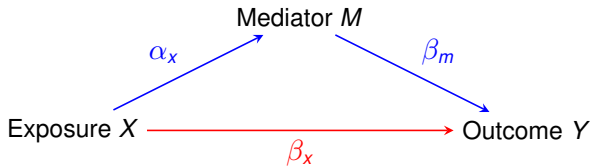
$$Y = \beta'_0 + \underbrace{\beta'_x}_{\text{total effect}} X + \epsilon'_y$$

$$M = \alpha_0 + \underbrace{\alpha_x}_{\text{effect of } X \text{ on } M} X + \epsilon_m$$

The indirect effect is commonly calculated as

$$\beta'_x - \beta_x = \alpha_x \beta_m$$

# PATH ANALYSIS / STRUCTURAL EQUATION MODELS



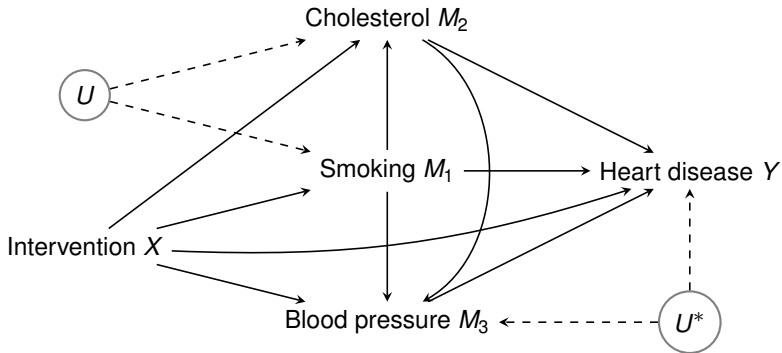
These results follow more from the broader theory on path analysis and structural equation models:

- Effects in sequence can be combined by multiplication;
- Effects in parallel can be combined by addition.

(Wright, 1934)

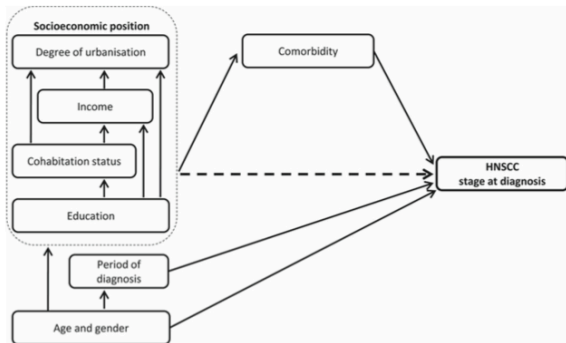
BUT...

Confounding is subtle, and not given due consideration.



# BUT...

Techniques for linear models need **not** carry over to non-linear models.



*‘Adjustment for comorbidity left the estimates virtually unchanged, suggesting that comorbidity does not play a major role in the socioeconomic differences in the stage distribution of HNSCC.’*

(Olsen et al., 2015)

# NON-COLLAPSIBILITY

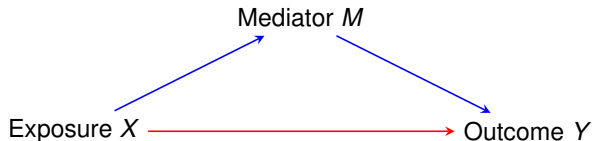
- This analysis is problematic.
- Adding even independent variables to a non-linear model tends to change coefficients systematically.



- This phenomenon is called **non-collapsibility**.
- This is perhaps best known in the context of logistic mixed effects models.
- It makes difference-of-coefficient methods problematic.

# INTERPRETATION

- Product-of-coefficient methods are also problematic.



- E.g. consider models

$$E(M|X) = \alpha_0 + \alpha_1 X$$

$$\text{logit}P(Y = 1|X, M) = \beta_0 + \beta_1 X + \beta_2 M$$

- *How to interpret the product  $\alpha_1 \beta_2$   
of a mean difference and a log odds ratio?*



# MEDIATION ANALYSIS 2.0

- The problem with traditional mediation analysis is that there is **an abundance of estimation methods, but no understanding what they are estimating.**
- In a revolutionary paper, Robins and Greenland (1992) identified these concerns and came up with direct and indirect effect **estimands.**
- This has led to a **complete re-development on mediation analysis,** which has taken off rapidly since 2010.

(Robins and Greenland, 1992; Pearl, 2001; Didelez et al., 2006; VanderWeele and Vansteelandt, 2009, 2010; Imai et al., 2010; VanderWeele, 2015)

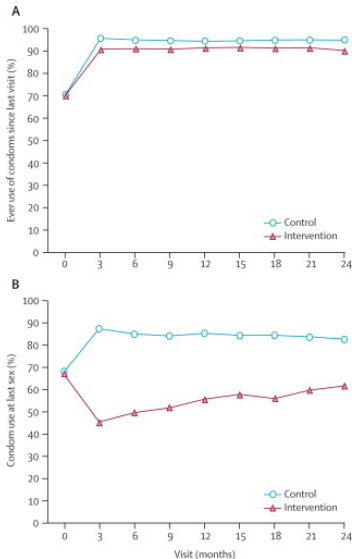
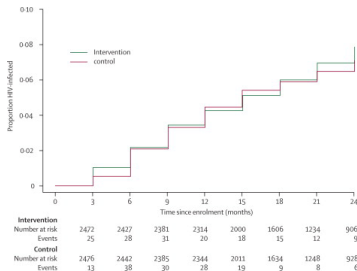
## Take home message

Modern mediation analysis techniques are **applicable to non-linear models** and **careful about problems of confounding.**

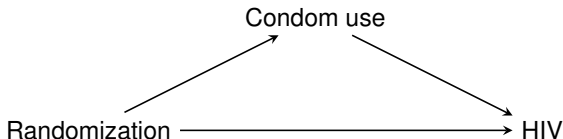
# NATURAL DIRECT AND INDIRECT EFFECTS

# THE MIRA TRIAL

Padian et al.,  
Lancet 2007



# HOW TO FORMALISE THE NOTION OF A DIRECT EFFECT?



- *What would have been the intervention effect on HIV incidence if indirect effects via condom use had been eliminated?*
- There are many different ways to formalise this.
- *What would be the intervention effect*
  - *if male condoms were never used?*  
(controlled direct effect)
  - *if intervention left each individual's condom use unchanged?*  
(natural direct effect)

SUPPOSE WE RANDOMISE A GIVEN WOMAN TO CONTROL...

Control

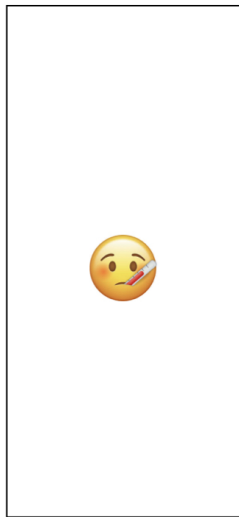


# SUPPOSE WE RANDOMISE THAT WOMAN TO INTERVENTION...

Control



Intervention



# WHAT IF SHE HAD NOT CHANGED CONDOM USE...?

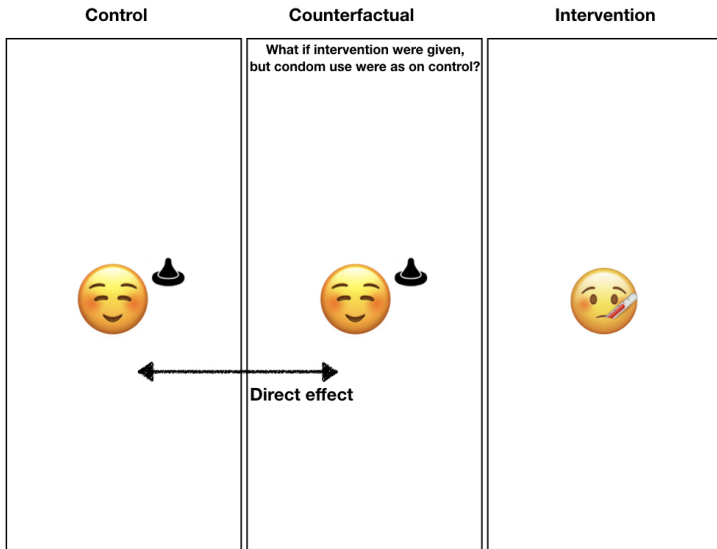
Control

Counterfactual

Intervention

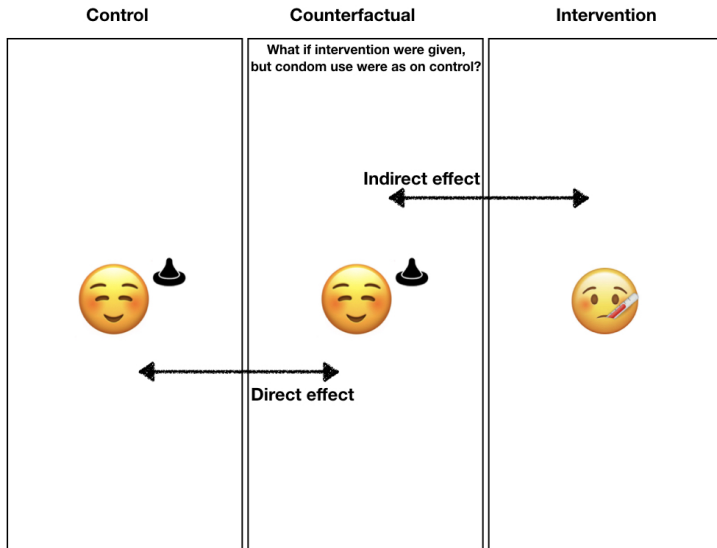
|                                                                                   |                                                                                                                                                                  |                                                                                     |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  | <p>What if intervention were given,<br/>but condom use were as on control?</p>  |  |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|

# NATURAL DIRECT EFFECT

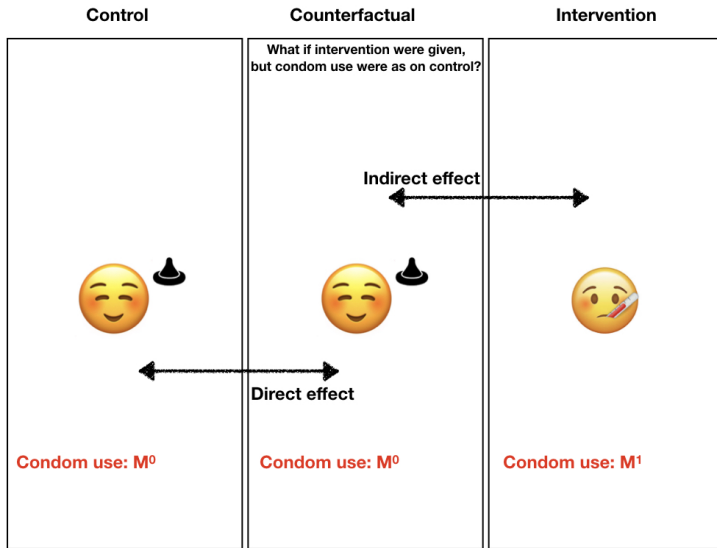




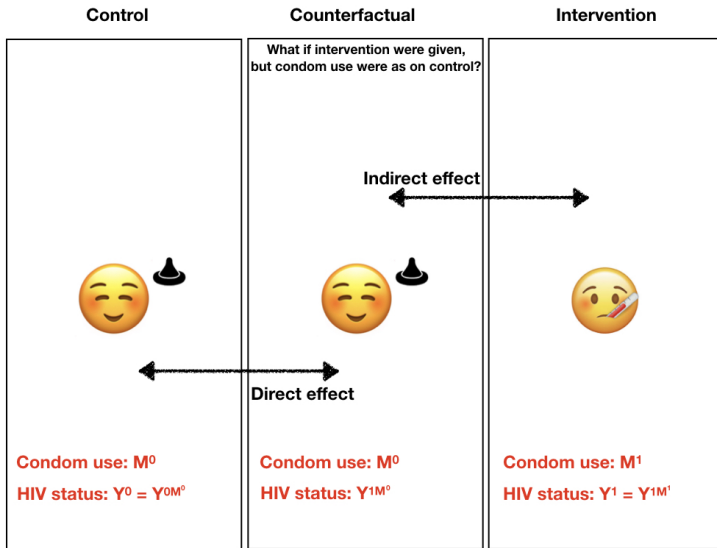
# NATURAL INDIRECT EFFECT



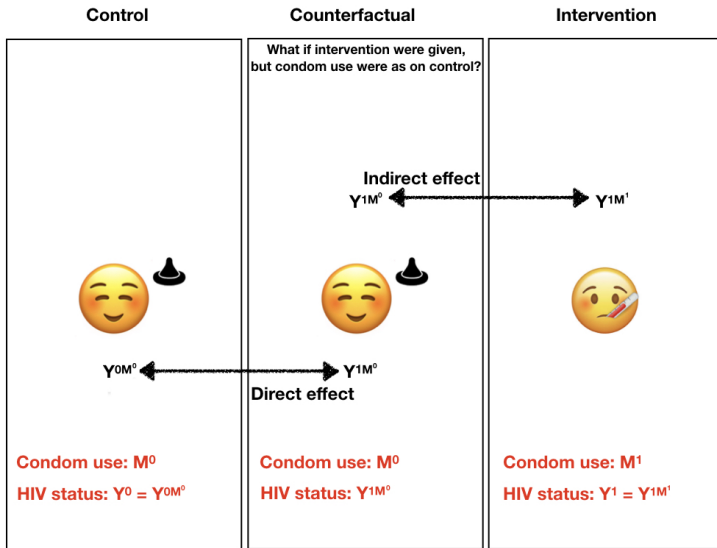
# COUNTERFACTUAL DATA (1)



## COUNTERFACTUAL DATA (2)



# FORMAL DEFINITION OF NATURAL (IN)DIRECT EFFECT



# POPULATION NATURAL (IN)DIRECT EFFECTS

- Natural direct effect:

$$E\left(Y^{1M^0} - Y^{0M^0}\right) \quad \text{or} \quad E\left(Y^{1M^0}\right) / E\left(Y^{0M^0}\right)$$

- Natural indirect effect:

$$E\left(Y^{1M^1} - Y^{1M^0}\right) \quad \text{or} \quad E\left(Y^{1M^1}\right) / E\left(Y^{1M^0}\right)$$

- Alternatively, we may control condom use  
at the level seen on intervention, e.g.:

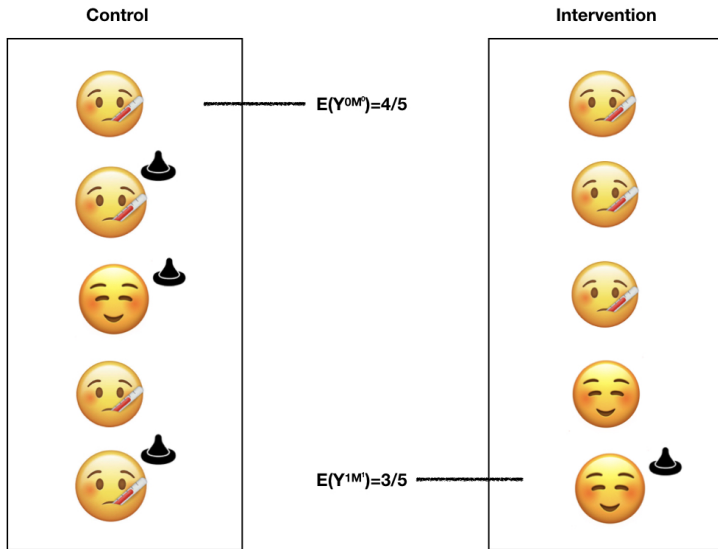
$$E\left(Y^{1M^1} - Y^{0M^1}\right)$$

and

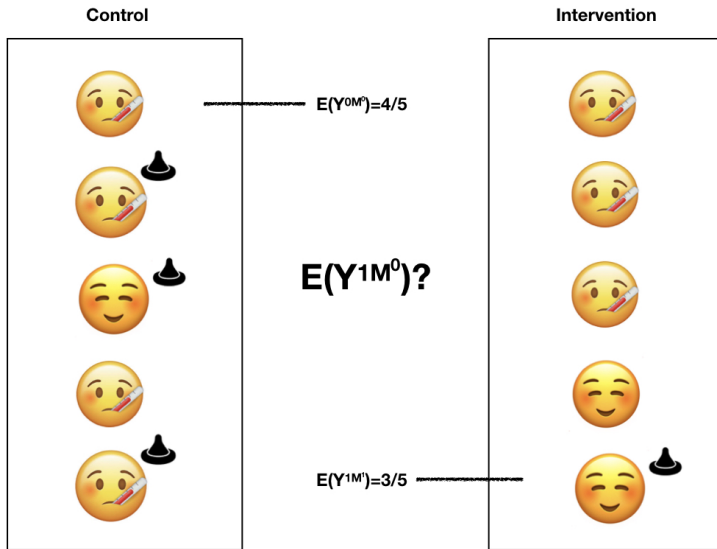
$$E\left(Y^{0M^1} - Y^{0M^0}\right)$$

# THE MEDIATION FORMULA

# CONSIDER NOW A 'REAL' STUDY



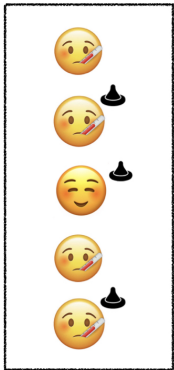
# THE FUNDAMENTAL PROBLEM IN MEDIATION ANALYSIS



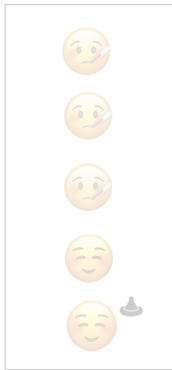


# HOW TO INFER $E(Y^{1M^0})$ ?

Control



Intervention

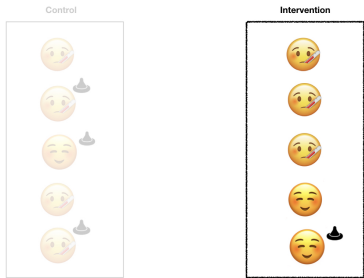


- Using the [mediation formula](#).

(Pearl, 2001; VanderWeele and Vansteelandt, 2009, 2010)

- We do not observe  $M^0$  for everyone.
- We will therefore ‘stochastically’ predict  $M^0$  using a prediction model for the ‘untreated’.
- This model should adjust for confounding of the  $X$ - $M$  association.

# HOW TO INFER $E(Y^{1M^0})$ ?



mediation formula:  $E(Y^{1M^0})$  equals

$$\int E(Y|X=1, m, c) f(m|X=0, c) f(c) dm dc$$

(Pearl, 2001)

- We next predict what outcome would be on treatment, at level  $M^0$ .
- And then average these predictions.
- This is very flexible!
- Since we evaluate the effect of 2 interventions, *setting treatment to 1 and mediator to  $M^0$* , the prediction model must **adjust for confounding of the  $X$ - $Y$  and  $M$ - $Y$  associations**.
- Mediation analyses thus necessitate confounding adjustment, even in randomised experiments!

# TOO GOOD TO BE TRUE?

- We can never observe  $Y^{1M^0}$ .
- So how come we found a way to identify  $E\left(Y^{1M^0}\right)$ ?
- It is because of implicit reliance on 4 **untestable assumptions**:

$$Y^{1m} \perp\!\!\!\perp X|C, \quad M^0 \perp\!\!\!\perp X|C, \quad Y^{1m} \perp\!\!\!\perp M|X=1, C, \quad Y^{1m} \perp\!\!\!\perp M^0|C$$

- The first 2 assumptions hold if  $X$  is randomly assigned (given  $C$ ).
- The third assumption states that the  $M$ - $Y$  association must be unconfounded, given 'baseline' variables  $C$ , thereby making extensions to multiple mediators subtle.

(VanderWeele T, Vansteelandt S. Mediation analysis with multiple mediators. Epidemiologic methods. 2014 Jan 3;2(1):95-115.)

- The last 'cross-world' assumption is closely related, but has been much criticized.

(Robins and Richardson, 2010)

- Because of it, even if one had experimental data to learn the effect of  $X$  on  $M$  and of  $M$  on  $Y$ , adjustment for  $C$  remains needed to combine these effects.

# DEBIASED MACHINE LEARNING

# THE MEDIATION FORMULA

- The mediation formula

$$\begin{aligned} E\left(Y^{1M^0}\right) &= \int E(Y|X=1, m, c) f(m|X=0, c) f(c) dm dc \\ &= E\left[E\{E(Y|X=1, M, C)|X=0, C\}\right] \end{aligned}$$

has become extremely popular and given rise to a variety of (semi-)parametric approaches, implemented in R-packages `mediation` and `medflex`.

(VanderWeele and Vansteelandt, 2009, 2010; Imai et al., 2010; Lange, Vansteelandt and Bekaert, 2012; VanderWeele, 2015; Steen et al., 2016, 2017)

- *How would you use this to construct an estimator?*

# THE INFLUENCE CURVE OF $E(Y^{1M^0})$

- Even so, it cannot readily be used for data-adaptive inference in view of plug-in bias.
- We will therefore work with its **influence curve**

$$\begin{aligned} & \frac{X}{P(X=1|C)} \frac{f(M|X=0, C)}{f(M|X=1, C)} \{Y - E(Y|X=1, M, C)\} \\ & + \frac{1-X}{P(X=0|C)} \left[ E(Y|X=1, M, C) - \int E(Y|X=1, m, C) f(m|X=0, C) dm \right] \\ & + \int E(Y|X=1, m, C) f(m|X=0, C) dm - E(Y^{1M^0}) \end{aligned}$$

- *How would you use this to construct an estimator and its standard error?*

## A MORE ATTRACTIVE PARAMETERISATION

- Using data-adaptive methods to model the density  $f(M|X, C)$  is tricky when - as often - the mediator is continuous.
- Even more so because of inverse density weighting.
- We therefore rewrite the first term using Bayes' rule as

$$\frac{X}{P(X=0|C)} \frac{P(X=0|M, C)}{P(X=1|M, C)} \{Y - E(Y|X=1, M, C)\}$$

where  $P(X=1|M, C)$  is easier to model.

# DEBIASED MACHINE LEARNING

- A debiased machine learning estimator of  $E(Y^{1M^0})$  is thus obtained as

$$\begin{aligned} & \frac{1}{n} \sum_{i=1}^n \underbrace{\frac{X_i}{\hat{P}(X_i = 0|C_i)}}_g \underbrace{\frac{\hat{P}(X_i = 0|M_i, C_i)}{\hat{P}(X_i = 1|M_i, C_i)}}_h \left\{ Y_i - \underbrace{\hat{E}(Y_i|X_i = 1, M_i, C_i)}_b \right\} \\ & + \frac{1 - X_i}{\hat{P}(X_i = 0|C_i)} \left[ \hat{E}(Y_i|X_i = 1, M_i, C_i) - \hat{E} \left\{ \hat{E}(Y_i|X_i = 1, M_i, C_i) | X_i = 0, C_i \right\} \right] \\ & + \underbrace{\hat{E} \left\{ \hat{E}(Y_i|X_i = 1, M_i, C_i) | X_i = 0, C_i \right\}}_u \end{aligned}$$

where the letters  $g, h, b, u$  refer to the names of the learners in the software - see later.

- Its standard error can be calculated like for a sample average, as if the nuisance parameters were known.
- Cross-fitting is recommended.



# CASE STUDY: JOBS II

# JOB SEARCH INTERVENTION STUDY (JOBS II)

(Imai et al., 2010)

- Randomized field experiment that investigates the efficacy of a job training intervention on unemployed workers.
- The program is designed to not only increase reemployment among the unemployed, but also enhance the mental health of the job seekers.
- 1801 unemployed workers received a pre-screening questionnaire.
- They were then **randomly assigned** to:
  - treatment: participating in job-skills workshops.
  - control: receipt of a booklet describing job-search tips.
- We will investigate how much of the treatment effect on depression scores is mediated by the level of job-search self-efficacy.
- For this, we will control for confounding by baseline variables  $W$

(economic hardship, baseline depression score sex, age, occupation, marital status, race, education and income)

# TMLE OF NATURAL DIRECT EFFECT

```
> install.packages(mediation)
> library(mediation)
> data(jobs)
> remotes::install_github("nhejazi/medoutcon")
> library(medoutcon)
> tmle_de <- medoutcon(W = cbind(jobs$econ_hard, jobs$depress1, jobs$sex, jobs$age, jobs$occp,
  jobs$marital, jobs$nonwhite, jobs$educ, jobs$income), A = jobs$treat, Z = NULL, M = jobs$job_seek,
  Y = jobs$depress2, effect = "direct", estimator = "tmle")
> tmle_id <- medoutcon(W = cbind(jobs$econ_hard, jobs$depress1, jobs$sex, jobs$age, jobs$occp,
  jobs$marital, jobs$nonwhite, jobs$educ, jobs$income), A = jobs$treat, Z = NULL, M = jobs$job_seek,
  Y = jobs$depress2, effect = "indirect", estimator = "tmle")
```

This uses 5-fold cross-fitting by default.

# TMLE OF NATURAL INDIRECT EFFECT

```
> tmle_de
```

Natural Direct Effect

Estimator: tmle

Estimate: -0.034

Std. Error: 0.041

95% CI: [-0.115, 0.047]

```
> tmle_ide
```

Natural Indirect Effect

Estimator: tmle

Estimate: -0.013

Std. Error: 0.01

95% CI: [-0.033, 0.007]

*Interpret the results.*

## USING SUPERLEARNER...

- The default learners are standard generalised linear models.
- To use SuperLearner, we must call the `sl3` library, which contains a richer class of learners.

```
> install_github("tlverse/sl3@devel")
> library(sl3)
> lrn_glm <- Lrnr_glm$new()
> lrn_ridge <- Lrnr_glmnet$new(alpha = 0)
> lrn_lasso <- Lrnr_glmnet$new(alpha = 1)
> lrn_ranger <- Lrnr_ranger$new()
> lrn_gam <- Lrnr_gam$new()
> stack <- Stack$new(lrn_glm, lrn_ridge, lrn_lasso, lrn_ranger, lrn_gam)
> sl <- Lrnr_sl$new(learners = stack, metalearner = Lrnr_nnls$new())
> tmle_de <- medoutcon(W = cbind(jobs$econ_hard, jobs$depress1, jobs$sex, jobs$age, jobs$occp,
  jobs$marital, jobs$nonwhite, jobs$educ, jobs$income), A = jobs$treat, Z = NULL, M = jobs$job_seek,
  Y = jobs$depress2, effect = "direct", estimator = "tmle",
  g_learners = sl, h_learners = sl, b_learners = sl)
# g_learners: P(X|C)
# h_learners: P(X|M,C)
# b_learners: E(Y|M,X,C)
```

# USING SUPERLEARNER...

```
> tmle_de
```

Natural Direct Effect

Estimator: tmle

Estimate: -0.041

Std. Error: 0.042

95% CI: [-0.122, 0.04]

```
> tmle_ide <- medoutcon(W = cbind(jobs$econ_hard,jobs$depress1,jobs$sex,jobs$age,jobs$occp,  
  jobs$marital,jobs$nonwhite,jobs$educ,jobs$income), A = jobs$treat, Z = NULL, M = jobs$job_seek,  
  Y = jobs$depress2, effect = "indirect", estimator = "tmle",  
  g_learners = sl, h_learners = sl, b_learners = sl)  
> tmle_ide
```

Natural Indirect Effect

Estimator: tmle

Estimate: -0.016

Std. Error: 0.01

95% CI: [-0.035, 0.003]